Experiment 6

End-to-End ML Pipeline Implementation

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Task 1 - Dataset Preparation

Dataset: Breast Cancer Wisconsin (Diagnostic), loaded from scikit-learn (569 records, 30 numeric features).

Target: ['malignant', 'benign'] (0 = malignant, 1 = benign). Target distribution: {'1': 357, '0': 212}.

Data quality: missing values = 0, duplicate rows removed = 0, final samples = 569.

All 30 features are numeric; no categorical encoding required. 80/20 stratified train/test split with random_state=42 for reproducibility.

Task 2 - Pipeline Design

Workflow: Data Loading → Cleaning / Prep → Feature Engineering → Model Training → Model Evaluation → Prediction.

Preprocessing is fitted only on training data via a scikit-learn Pipeline (SimpleImputer → StandardScaler → RandomForestClassifier) and consistently applied to test data.

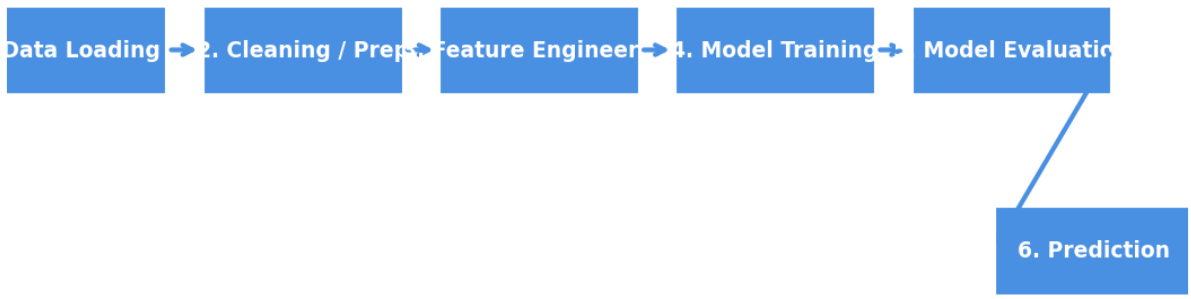


Figure 1: End-to-end ML pipeline architecture diagram

Task 3 - Modular Pipeline Implementation

The workflow is split into reusable Python modules under **src/**:

Module	Responsibility
data_loader.py	Load dataset, inspect data quality, save inspection summary
preprocessing.py	Handle missing values/duplicates, train/test split
feature_engineering.py	Validate dtypes, align train/test features, record transformations

train.py	Build scikit-learn Pipeline, fit model, compute metrics, save model artifact
evaluate.py	Compute Accuracy/Precision/Recall/F1/ROC-AUC, compare runs for reproducibility
predict.py	Load saved pipeline, generate predictions with confidence, save to CSV
main.py	Orchestrate all stages end-to-end; reproducibility check

Configuration is managed in **config/config.yaml** (no hard-coded paths). Model and preprocessing components are saved together as **models/model.joblib**.

Task 4 - Build and Execute the Pipeline

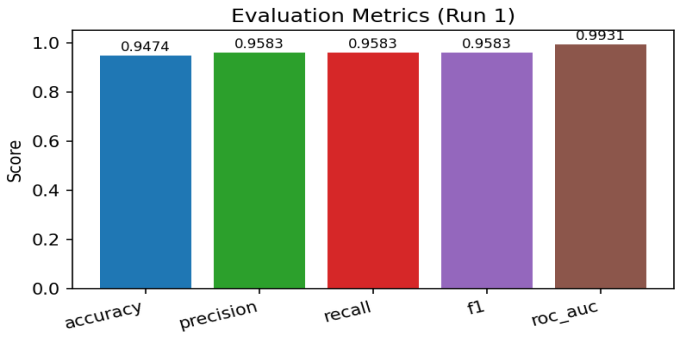
Executed via a single command: python src/main.py. The pipeline runs end-to-end without manual notebook cells.

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End-to-End Pipeline Execution

PS C:\batch_project\exp-6> python src/main.py
2026-10-22 18:03:59,135 | INFO | Configuration loaded from config/config.yaml
2026-10-22 18:03:59,135 | INFO | === Reproducibility Run 1 ===
2026-10-22 18:03:59,158 | INFO | Task 1: Dataset Preparation
2026-10-22 18:03:59,158 | INFO | Loaded dataset: 569 samples, 30 features
2026-10-22 18:03:59,213 | INFO | Train/test split: (455, 30) / (114, 30)
2026-10-22 18:03:59,219 | INFO | Model Training
2026-10-22 18:03:59,719 | INFO | Evaluation metrics: accuracy=0.9474, f1=0.9583, roc_auc=0.9931
2026-10-22 18:03:59,795 | INFO | Saved model to models/model.joblib
2026-10-22 18:03:59,887 | INFO | Saved 114 predictions to outputs/run_1_predictions.csv
2026-10-22 18:03:59,889 | INFO | === Reproducibility Run 2 ===
2026-10-22 18:04:00,672 | INFO | Evaluation metrics: accuracy=0.9474, f1=0.9583, roc_auc=0.9931
2026-10-22 18:04:00,880 | INFO | Overall consistent: True
2026-10-22 18:04:00,882 | INFO | End-to-end Experiment 6 pipeline complete.
```

Figure 2: Terminal output of pipeline execution

Evaluation metrics: {"accuracy": 0.9473684210526315, "precision": 0.9583333333333334, "recall": 0.9583333333333334, "f1": 0.9583333333333334, "roc_auc": 0.9930555555555556}.



Task 5 - Reproducibility and Prediction

The pipeline was executed twice (run_1 and run_2) with identical configuration and random_state=42. All metrics match exactly (difference = 0.0).

Metric	Run 1	Run 2	Difference	Consistent
accuracy	0.947368	0.947368	0.000000	Yes
precision	0.958333	0.958333	0.000000	Yes
recall	0.958333	0.958333	0.000000	Yes
f1	0.958333	0.958333	0.000000	Yes

roc_auc	0.993056	0.993056	0.000000	Yes
Overall	-	-	-	Yes

Prediction stage loaded the saved model/pipeline and predicted 114 unseen test records using identical preprocessing as training.

Task 6 - Documentation and Report

Project structure:

exp-6/
data/ | src/ | models/ | outputs/ | config/
src/data_loader.py, preprocessing.py, feature_engineering.py, train.py, evaluate.py, predict.py, main.py
config/config.yaml | models/model.joblib | outputs/*.json, outputs/*.csv

Dataset: 569 samples, 30 features. Model: RandomForestPipeline (200 trees). Preprocessing: median impute + standard scale.

Output format: predictions CSV with record_id, predicted_class, predicted_label, confidence, timestamp, model_version. Sample output rows:

record_id	predicted_class	predicted_label	confidence	model_version
REC0001	0	malignant	1.0	v1.0
REC0002	1	benign	1.0	v1.0

Observations: Pipeline executed successfully end-to-end. Reproducibility verified (overall_consistent: True). No errors encountered. All metrics consistent across repeated runs.

Conclusion: A modular, reproducible end-to-end ML pipeline was implemented per Experiment 6 specifications, covering data preparation through prediction with full documentation.